

Eunha Lee

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Website	https://eunha-l.github.io/
Research Interests	**Drug Discovery · Biomolecular Modeling Protein Interactions**

ABOUT ME

I am an M.S. student in the Computer Science & Engineering Department at Korea University and a member of **DMIS Lab**, advised by **Prof. Jaewoo Kang**. My research focuses on **AI-driven drug discovery**, spanning **biomolecular modeling** and **protein interactions**. I am particularly interested in antigen–antibody interactions and developing predictive and generative models that link molecular structure to biological function.

EDUCATION

M.S. in Computer Science and Engineering, Korea University Sep 2025 – Present

DMIS Lab (Advisor: Prof. Jaewoo Kang)

B.S. in Biosystems and Medical Sciences, Korea University Mar 2021 – Aug 2025

GPA: 4.06 / 4.5

PUBLICATIONS

Eunha Lee*, Seungheun Baek*, Hajung Kim, Jaewoo Kang.
"EpiPose: Antibody-Specific Epitope Prediction from Unbound Structures by Learning to Reweight Docking Poses." *Under review*. **[paper]**

- Predicting antibody-specific epitopes from unbound structures while accounting for binding-induced structural changes.
- Devised the overall pipeline and conducted experiments with structure prediction models (e.g., AlphaFold 3, Boltz-2, Ibex) and protein language models (e.g., ESM, AntiBERTy).

Hajung Kim*, Junseok Choe*, **Eunha Lee**, Sohyun Chung.
"CycloFlow: Internal Coordinate Flow Matching and Geometric Ring Closure for Cyclic Peptide Conformer Generation." *Under review*. **[paper]**

- Generating 3D conformational ensembles of cyclic peptides using flow matching and geometric ring closure.
- Reviewed conformer generation methods and contributed to model design discussions.

Seungheun Baek, **Eunha Lee**, Yan Ting Chok.
"FreeDIA: Search-Free Pretraining for DIA de novo Peptide Sequencing." *Under review*. **[paper]**

- De novo peptide sequencing from DIA-MS/MS spectra without sequence database searches.
- Preprocessed MS/MS data, reimplemented baseline models, and performed BLAST-based sequence analysis and background research on PTMs.

Hajung Kim, **Eunha Lee**, Sohyun Chung, Jueon Park, Seungheun Baek, Jaewoo Kang. **"ATTNSOM: Learning Cross-Isoform Attention for Cytochrome P450 Site-of-Metabolism."** *ISMB 2026*. **[paper]**

- Predicting drug metabolism sites by learning shared patterns across cytochrome P450 isoforms.
- Implemented and evaluated deep learning models using PyTorch and RDKit, including baseline reproduction and ablation studies.

EXPERIENCE

AIGEN Sciences Inc. **[Link]**

AI Research Intern Feb 2026 – May 2026

- Conducted ADMET prediction research on cytochrome P450 inhibition and derived 3D interaction features from predicted protein–ligand complex structures using PLIP.

Undergraduate Researcher Jan 2025 – Jul 2025

- Analyzed biological databases (e.g., DepMap, ChEMBL) to evaluate LLM-generated hypotheses.
- Analyzed clinical data from cancer patients and conducted baseline experiments for survival analysis.

DMIS Lab, Korea University

AI Researcher Sep 2025 – Feb 2026

- ARPA-H project: LC–MS/MS-based Biomarker Discovery for Multiple Cancers
- Preprocessed large-scale patient LC–MS/MS data and identified data quality issues.

Disease Study Society (DSS), Korea University **[Link]**

Member Mar 2024 – Dec 2024

- On-campus academic society. Discussed and presented biology research papers.

Aging, Metabolism and Physiology Lab, DGIST **[Link]**

Undergraduate Intern Dec 2023 – Feb 2024

- Worked in a lab studying metabolism and immune physiology.
- Performed wet lab experiments, including Western blotting, PCR, and immunostaining.

COMPETITIONS

ATEC2026, *Global AI and Robotics Competition* **[Link]** Jul 2026 – Sep 2026

- An embodied AI competition focused on real-world robotic tasks. Our team placed in the top 10 of 93 teams in the online qualifier and advanced to the offline round in Hong Kong.
- Implemented locomotion code with reinforcement learning and operated the real robot as the team representative.